

Acid-Base Indicators

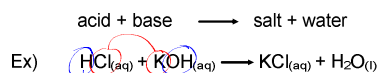
An indicator is a substance that changes colour depending on whether it is in an acid or base

Indicator	Colour in Acid	Colour in Base
bromethymol blue	yellow	blue
phenolphthalein	colourless	pink
phenol red	yellow	red/pink
litmus	red	blue
methyl orange	red	orange/yellow

Some *natural indicators* include, strawberries, red cabbage, tea, blueberries, beets, cranberries (cranberry juice), grape juice, geranium leaves

Acid-Base Neutralization

Recall from unit 2, neutralization reactions are a type of double displacement reaction:



Neutralization is the exact point at which the number of moles of H_3O^+ ions equals the moles of OH^- ions.

Recall: $C = \frac{n}{V} \longrightarrow n = CV$

Therefore: $n_{\text{acid}} = n_{\text{base}}$
or
 $C_A V_A = C_B V_B$

Ex) What is the volume of 0.50 M HCl required to neutralize 10.0 mL of 1.0 M NaOH?

$$C_A V_A = C_B V_B$$

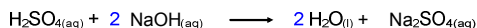
$$(0.50) V_A = (1.0)(10.0)$$

$$V_A = \frac{(1.0)(10.0)}{(0.50)}$$

$$= 20$$

$$= 2.0 \times 10 \text{ mL}$$

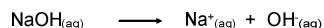
Consider the neutralization reaction between H_2SO_4 and NaOH:



When H_2SO_4 dissociates it produces 2 H^+ ions.



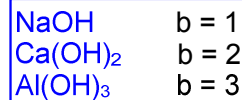
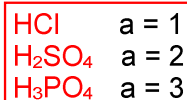
NaOH only produces 1 OH^- ion.



This is not a 1:1 ratio like we have seen previously. We need to adjust our neutralization equation accordingly:

$$aC_A V_A = bC_B V_B$$

a - the number of H^+ ions produced when the acid dissociates
b - the number of OH^- ions produced when the base dissociates



Ex) What volume of 1.5 M H_2SO_4 is required to neutralize 45 mL of 0.60 M NaOH?

$$a C_A V_A = b C_B V_B$$

$$(2)(1.5) V_A = (1)(0.60)(45)$$

$$V_A = 9.0 \text{ mL}$$

Ex) 13.2 mL of 0.125 M H_3PO_4 was used to neutralize 15.2 mL of $\text{Ca}(\text{OH})_2$. What is the concentration of the $\text{Ca}(\text{OH})_2$?

$$a C_A V_A = b C_B V_B$$

$$(3)(0.125)(13.2) = (2) C_B (15.2)$$

$$4.95 = C_B 30.4$$

$$0.163 \text{ M} = C_B$$

Ex) Stomach acid (HCl) has a concentration of approximately 0.10M. How much stomach acid can be neutralized by taking an antacid tablet that contains 100.0 mg of aluminum hydroxide, $\text{Al}(\text{OH})_3$?

$$n_{\text{Al}(\text{OH})_3} = \frac{0.1000}{78.01}$$

$$= 0.001282 \text{ mol}$$

$$n = CV$$

$$a C_A V_A = b C_B V_B$$

$$a C_A V_A = b n_B$$

$$(1)(0.10) V_A = 3(0.001282)$$

$$V_A = 0.03846$$

$$V_A = 0.038 \text{ L}$$

or
38 mL